

COURSE TITLE : POLYMER PROCESSING & MACHINERY
COURSE CODE : 4093
COURSE CATEGORY : A
PERIODS/WEEK : 6
PERIODS/SEMESTER: 84/4
CREDITS : 5

TIME SCHEDULE

Module	Topics	Period
1	Mixing machinery	21
2	Moulding Processes and Machinery	21
3	Extruders	21
4	Calendars and Thermoforming	21
TOTAL		84

COURSE OUTCOME

Sl.	GO	Student will be able to
1	1	To comprehend the mixing machinery
2	2	To comprehend moulding Process and machinery
3	3	To understand the working principles , parts and accessories of extruders
4	4	To understand the working principles , parts and accessories of calendars
	5	To understand the working principles and process of thermoforming

SPECIFIC OUTCOME

MODULE I THE MIXING MACHINERY

1.1.0 To comprehend the mixing machinery

- 1.1.1. Classify the mixing equipments
- 1.1.2. Explain the working principle, various parts and accessories, safety devices and important uses of Open Two Roll Mixing Mill.
- 1.1.3. Explain the working principle, various parts and accessories of Banbury and Intermix.
- 1.1.4. Describe fill factor, mixing sequences, distributive mixing, dispersive mixing and upside down mixing.
- 1.1.5. Compare and contrast between Open two roll mill and internal mixers.
- 1.1.6. Distinguish between Banbury and Intermix.
- 1.1.7. Explain the temperature and pressure control systems of mixing equipments.

- 1.1.8. Discuss various continuous mixing equipments like Double R mixer, Farrel continuous mixer, Transfer mixer and MVX mixer.

MODULE II THE MOULDING MACHINERY

2.1.0 To comprehend moulding machinery.

- 2.1.1. Define compression moulding, Transfer moulding and injection moulding.
- 2.1.2. Explain compression moulding process using flash type, positive and semi positive moulds.
- 2.1.3. Explain the working principle, different parts and accessories of Hydraulic Press with neat sketches.
- 2.1.4. Describe the process of Transfer moulding.
- 2.1.5. Compare the merits and demerits of compression and transfer moulding.
- 2.1.6. Explain the different parts and accessories of reciprocating and screw type Injection moulding machine with neat sketches.
- 2.1.7. Describe the injection moulds, different clamping systems and ejection mechanism of Injection moulding process.
- 2.1.8. Identify the trouble shooting in IM, TM and CM process.
- 2.1.9. Explain RIM and RRIM.
- 2.1.10. Describe rotational moulding.

MODULE III EXTRUDERS

3.1.0 To understand the working principles , parts and accessories of extruders

- 3.1.1. Define Extrusion process.
- 3.1.2. Classify the extruders.
- 3.1.3. Explain the different parts and accessories of Ram and screw extruders with sketches.
- 3.1.4. Describe cold feed , Hot feed, T-head, Dual head, Twin screw, pin and vented extruders
- 3.1.5. Describe pipe extrusion, wire and cable extrusion.
- 3.1.6. Explain the blown film and flat film extrusion.
- 3.1.7 Explain the temperature control systems of extruders.
- 3.1.8. Give the sizes and specifications of extruders.
- 3.1.9. Describe blow moulding process.

MODULE IV CALENDARS AND THERMOFORMING

4.1.0 To understand the working principles , parts and accessories of calendars

- 4.1.1. Define calendaring process.
- 4.1.2. Give the different calendar configurations.
- 4.1.3. Explain the different part, and accessories of calendars.
- 4.1.4. Define profile calendaring.
- 4.1.5. Compare calendars with profile calendars.
- 4.1.6. Explain gauge compensation techniques like roll cambering (crowning), roll crossing and roll bending.
- 4.1.7. Explain spreading process.
- 4.1.8. Explain the parts, working and features of spreading machine.
- 4.1.9. Define thermoforming.

- 4.1.10. Explain the principle and process of vacuum forming, pressure forming and plug assisted forming.
- 4.1.11. Give examples of thermoformed products.

CONTENT DETAILS

MODULE-I

Mixing machinery

Classification- two roll mixing mill- working principle, parts, accessories, safety devices, uses. Banbury- working principle, parts, accessories, safety devices. Intermix- working principle, parts. Fill factor, mixing sequences, distributive mixing, dispersive mixing, upside down mixing.

MODULE-II

Moulding

Compression moulding, process, moulds-flash, positive, semi positive. Transfer moulding- process, machinery, sketch. Injection moulding- process, comparison among three, Injection moulding machine- parts, accessories, working. Hydraulic press- working principle, parts, accessories. Injection moulds-clamping system-ejection mechanism. Trouble shooting- i/m, t/m, c/m. RIM- process, RRIM- process, advantages. Rotational moulding- process.

MODULE-III

Extrusion

Definition, process. Extruder- classification- cold feed, hot feed, T-head, dual, twin screw, vented. Extruder-ram, screw- parts, accessories, sketches, pipe reextrusion, wire and cable, blown film extrusion, flat film extrusion. Extruder- size, specification, temperature control. Blow moulding- extrusion blow moulding, injection blow moulding.

MODULE-IV

Calendars and Thermoforming.

Calendar- configuration, parts, accessories, sketches, profile calendar- examples. Roll deflection and compensation technique like roll cambering (crowning) roll crossing and roll bending. Temperature control system of extruders and calendars. Thermoforming- vacuum forming, pressure forming, and plug assisted forming- products- examples.

REFERENCE BOOKS

- 1 Rubber Technology and manufacture-C.M. Blow (Publisher – Butterworth Hainemann Ltd, 1982)
2. Rubber processing and production organization -Philip K Freakley (Publisher: Springer 2011)
3. Blow moulding -David A Jenners
- 4 Injection moulding -Irvin Ruhin
5. Plastic Technology Hand book - William J Patton (Publisher - Reston Publishing Co. 1986)